

Practice Quiz: Agents

Part A: Multiple Choice

1. The module describes different crochet stitch types as analogous to different activation functions in neural networks. A **single crochet** (compact, dense, short connection) is compared to a step function, while a **double crochet** (taller, more open connection) is compared to a ReLU. The key idea is that: A) Taller stitches use more yarn, just as complex functions use more computation B) Each stitch type transforms the input (previous row) into output (current row) in a structurally distinct way C) Single crochet is always better than double crochet, just as step functions are always better than ReLU D) Stitch type does not matter as long as the row count is correct
2. When two crocheters follow the same pattern but produce fabric of different densities and drape, the variable that differs between them — analogous to neural network weights — is: A) The number of stitches per row B) The color of the yarn C) Their tension — how tightly they hold and work the yarn D) The direction they crochet (left-handed vs right-handed)
3. A crocheter making a gauge swatch and adjusting their hook size until the stitch count matches the pattern is performing a process most analogous to: A) Neural network architecture search B) Neural network weight training (gradient descent) C) Dropout regularization D) Data augmentation
4. At the moment of inserting the hook for the next stitch, the crocheter is at a **decision node**. The Active Inference cycle at this node involves: A) Prediction, sensation, prediction error, and update — all in a fraction of a second B) Reading the pattern once and never checking again C) Randomly choosing a stitch type to maximize variety D) Waiting for external instructions before each stitch
5. A crocheter working freeform (without a written pattern) is generating stitch decisions from: A) No model at all — pure randomness B) Their internal generative model — predictions about what will look and feel right C) Muscle memory only, with no cognitive involvement D) A hidden pattern that all crocheters share

6. In a crochet circle, multiple crafters each make a granny square that will be joined into a blanket. This is analogous to: A) A single monolithic neural network B) A modular neural network architecture — separate subnetworks connected into a larger structure C) A network with no connections between layers D) A random graph with no organizing principle
7. The recursive observation that "a neural network (the crocheter's brain) is building another neural network (the fabric)" highlights: A) That crochet is impractical and should be replaced by machine knitting B) That agency in Active Inference involves systems made of networks constructing networks C) That the fabric is conscious D) That only neuroscientists should crochet

Part B: Short Answer

1. Describe the Active Inference cycle that occurs at a single decision node — one stitch insertion — in a crocheter's work. Identify the generative model, the prediction, the sensory input, the prediction error, and the possible responses (active inference vs perceptual inference). Use a concrete example, such as a crocheter encountering unexpected resistance when inserting the hook.
2. Explain the analogy between a crochet pattern and a neural network specification. What architectural information does the pattern encode (layer count, nodes per layer, connection types, etc.)? What aspects of the "network" does the pattern leave to the crocheter's discretion, similar to how a network spec might leave weight initialization to the training process?
3. A beginning crocheter and an expert sit side by side, both making the same hat pattern. Compare their Active Inference processes. Where does the beginner spend most of their free energy? What has the expert's generative model learned that the beginner's has not? How does the concept of "weight training" (tension practice) explain the difference in their outputs?